WE-Dream: Generalizable Humanoid World-Action Models with Latent-Action Pretraining and Routed Attention



Fig. 1. **Overview of WE-Dream.** Human-video latent-action pretraining and robot post-training transfer experience to whole-body humanoid control; routed attention supports mixed-routing training and action-only inference. The robot dataset contains 4.02 hours across nine task families and 53 instruction-defined subtasks, with equal sectors denoting task families rather than duration and the room image compositing separately photographed demonstrations. The evaluation benchmark distinguishes familiar tasks (ID), changed positions, objects, and backgrounds (L1), and new skill compositions (L2).

***Abstract*— We introduce WE-Dream, a 16B-parameter humanoid world-action foundation model that transfers experience from up to 10k hours of human egocentric video to whole-body control through latent-action pretraining. We equip WE-Dream with routed attention to enable efficient action-only inference without generating future video. To evaluate both task acquisition and generalization, we introduce a room-scale real-robot benchmark built around nine long-horizon household tasks. We jointly post-train WE-Dream on these tasks to obtain a single generalist policy for Unitree G1, and evaluate its in-distribution performance (ID), generalization to new positions, objects, and backgrounds (L1), and ability to execute unseen compositions of learned skills (L2). On this benchmark, WE-Dream outperforms the strongest external baseline in each setting by 42.2, 51.7, and 65.0 percentage points on ID, L1, and L2, respectively. Scaling latent-action pretraining from 2k to 10k hours improves downstream performance. On the same 2k-hour dataset, latent-action pretraining yields stronger humanoid transfer than hand-pose supervision and action-free video pretraining, improving success over hand-pose supervision by 20.0, 33.3, and 40.0 percentage points on ID, L1, and L2, respectively. In a separate ID ablation, efficient action-only inference achieves similar aggregate success to joint video-action inference within the same mixed-routing checkpoint, while mixed training improves both modes over their fixed-route counterparts. With FP8 quantization and real-time chunking, action-only inference enables the 16B-parameter policy to run on a single RTX 5090 with $\sim$260 ms latency ($2.9\times$ speedup over joint mode), while maintaining a continuous 30 Hz whole-body action stream.**

I. Introduction

A household humanoid must coordinate manipulation with walking, turning, and changes in body posture to carry out everyday tasks. It must also adapt to unfamiliar objects and environments, and combine familiar skills according to new instructions. Building on video-generation backbones, world-action models (WAMs) offer a foundation for robot policies that generalize to new tasks and environments [1], [2]. Collecting robot demonstrations for every combination of skill, object, and scene is costly. Human egocentric video captures everyday interactions without requiring robot teleoperation, offering a scalable source of transferable interaction knowledge for pretraining humanoid WAMs [3]–[5].

The first question is which action representation best supports scaling humanoid WAMs with egocentric video. Prior human-to-robot learning methods use human wrist and hand poses as action supervision [4]–[6]. Tracked hand poses provide explicit geometry, but human and robot hands differ in size, joint structure, and degrees of freedom. Latent actions inferred from visual transitions [7]–[10] capture coarse-grained interaction semantics without prescribing human hand kinematics, leaving embodiment-specific control to robot post-training. We compare hand-pose, latent-action, and action-free video objectives on the same 2k-hour dataset in a pretraining ablation, alongside a baseline without human-video pretraining. In this ablation,

latent-action pretraining yields the strongest transfer on both demonstrated tasks and unseen skill compositions. We further expand the latent-action pretraining dataset to 5k and 10k hours to examine how performance scales with human experience. The 2k, 5k, and 10k models achieve ID success rates of 81.1%, 86.7%, and 90.0%, respectively.

Deploying humanoid WAMs requires efficient inference. Predicting video provides dense supervision during training, but generating future video increases inference cost. *The second question is whether WAMs still need to generate future video at test time.* Recent action-centered WAMs retain video supervision while omitting future-video generation at inference [11], [12]. We propose routed attention to mix joint and action-only attention routes during training and support both inference modes within a single model. In an ID ablation, action-only inference achieves similar aggregate success to joint video-action inference, while mixed-routing training improves both modes over their fixed-route counterparts.

Beyond demonstrated tasks, a policy must handle new conditions and new combinations of learned skills. To assess both task acquisition and generalization, we introduce a room-scale benchmark that evaluates a real humanoid at three levels. In-distribution (ID) evaluation measures acquisition of the demonstrated tasks. L1 evaluates generalization to new object positions, identities, or backgrounds while preserving task structure. L2 requires unseen combinations of familiar skills learned across the training tasks. For example, retrieving a bottle from a cabinet and disposing of it in a pedal-operated trash can combines cabinet access, lateral bottle grasping from drawer storage, and foot-actuated bin opening from rubbish disposal. Such tasks challenge instruction following by requiring the policy to select and sequence these skills according to a new instruction.

We introduce WE-Dream, a 16B-parameter humanoid world-action foundation model that combines latent-action pretraining with routed attention. We pretrain the model on up to 10k hours of human egocentric video and jointly post-train it on nine long-horizon household tasks to obtain a single language-conditioned policy for Unitree G1.

Our study makes three contributions:

- **Scaling humanoid WAMs from egocentric video via latent actions.** We identify latent-action prediction as the most effective pretraining objective for humanoid transfer through comparisons with hand-pose supervision, action-free video pretraining, and a baseline without human-video pretraining. At the 2k-hour pretraining scale, latent-action pretraining improves success over hand-pose supervision by 32.7%, 90.9%, and 177.8% relative on ID, L1, and L2, respectively. Scaling latent-action pretraining to 2k, 5k, and 10k hours yields ID success rates of 81.1%, 86.7%, and 90.0%, respectively.
- **Enabling efficient whole-body control with routed attention.** We propose routed attention, enabling joint and action-only paths to be mixed during training and compared at inference within a shared policy. In an ID ablation, efficient action-only inference achieves similar aggregate success to joint video-action inference from the same checkpoint. Moreover, mixed-routing training improves both inference modes over their corresponding fixed-route models. Combining action-only inference with FP8 quantization and real-time chunking [13], we deploy the 16B-parameter policy on a single RTX 5090 with $\sim$260 ms latency, a $2.9\times$ speedup over joint mode, and continuous 30 Hz whole-body action execution.
- **Benchmarking humanoid in-distribution performance and generalization.** We introduce a real-humanoid benchmark that evaluates in-distribution performance (ID), generalization to new positions, objects, and backgrounds (L1), and the ability to execute unseen compositions of learned skills (L2) within one multi-task setting. WE-Dream improves success rates by 42.2, 51.7, and 65.0 percentage points over the strongest external baseline in each of ID, L1, and L2, respectively.

II. Related Work

A. Human Egocentric Video for Robot Learning

Human video broadens robot learning beyond teleoperated demonstrations, but leaves a choice of how to represent the demonstrated actions. EgoVLA, Ψ_0, and EgoScale use explicit human motion targets to transfer egocentric experience to robot policies [4]–[6]. Latent action models instead infer compact transition codes from video [7]–[9]: LAPA uses them for policy pretraining, while DreamDojo scales latent-action conditioning for robot world-model. We study which action representation best supports scaling humanoid WAMs with human video. We compare hand-pose, latent-action, and action-free objectives on the same human-video data, then scale the latent-action recipe and evaluate transfer to humanoid whole-body skills and unseen skill compositions.

B. Training and Inference in World–Action Models

World–action models connect video-generative priors to control by jointly predicting future observations and actions [1], [2]. Motus supports multiple video–action prediction modes within a unified model [10]. Fast-WAM and GigaWorld-Policy retain video supervision while allowing action inference without future-video generation [11], [12]; Fast-WAM also compares controlled variants to examine the roles of video co-training and test-time imagination. Building on these advances, we ask how joint and action-only prediction should be combined during training and used at inference. Our routed attention mixes joint and action-only routes during training. We compare both inference modes within the same checkpoint and evaluate mixed-routing training against fixed-route training.

III. Method

A. Overview

Given a language instruction ℓ, an observation history $o_{\leq t}$ comprising the current and retained past head-camera observations, and proprioceptive state $\mathbf{s}_t$, our policy predicts a whole-body action chunk $\mathbf{a}_{t:t+H} = (\mathbf{a}_t, \ldots, \mathbf{a}_{t+H-1})$. We first pretrain on human egocentric video with latent-action

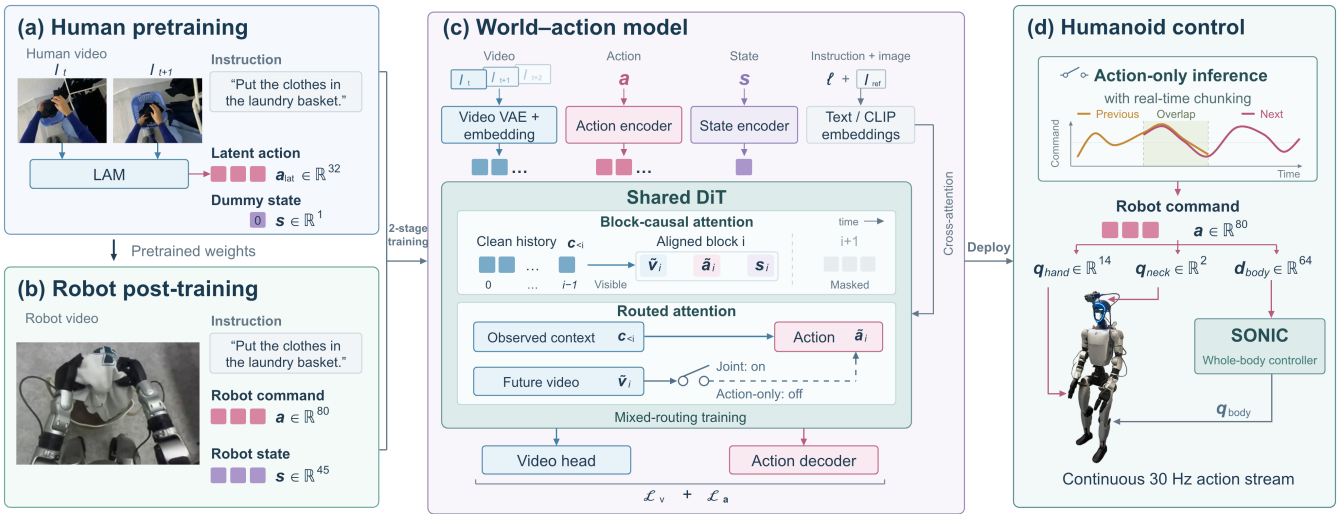


Fig. 2. **Model architecture, training, and humanoid control.** (a) Human-video pretraining uses latent-action targets extracted by a frozen latent action model (LAM). (b) Robot post-training adapts the pretrained model to humanoid commands and states. (c) Both stages use a shared DiT with block-causal attention. Routed attention controls action access to future-video tokens, supporting mixed-routing training with video and action flow-matching losses. Routing arrows indicate information flow to action queries; see Fig. 3 for detailed masks. (d) Action-only inference with real-time chunking supplies a continuous 30 Hz command stream. SONIC converts latent body commands into body joint targets, while hand and neck targets are executed directly.

targets, then adapt the pretrained model to the humanoid state and action spaces through multi-task robot post-training.

Initialized from Wan2.1-I2V-14B [14], the model shares a world–action backbone across both stages. We equip this backbone with routed attention to control whether action queries attend to future-video (*joint*) or not (*action-only*) during training and inference. Fig. 2 summarizes the model architecture, two-stage training, and deployment system.

B. Shared World–Action Backbone

1) Shared computation: As shown in Fig. 2 (c), video frames are encoded by a frozen VAE [14] and embedded into tokens, while action and state vectors use separate trainable input encoders. These interfaces map all three modalities to the same hidden dimension for processing by a shared DiT. Frozen UMT5-XXL [15] and CLIP [16] encode the instruction ℓ and reference image I_{ref}, respectively, for cross-attention conditioning. After the shared blocks, a video head and an action decoder predict the respective flow velocities.

2) Temporal organization: For each video sequence, we retain the first frame's VAE latent as a reference and divide the remaining latent frames into temporal blocks. Each block contains video latents, the corresponding action chunk, and one state token. During teacher-forcing training, block-causal attention allows the model to denoise each block's video and actions using preceding clean observations as context [1]. Let $\mathbf{c}_{<i}$ denote the clean video tokens preceding block i, $\tilde{\mathbf{v}}_i$ and $\tilde{\mathbf{a}}_i$ its noisy video and action tokens, and $\mathbf{s}_i$ its clean state token. Video queries read $\{\mathbf{c}_{<i}, \tilde{\mathbf{v}}_i, \tilde{\mathbf{a}}_i, \mathbf{s}_i\}$. Action queries always read $\{\mathbf{c}_{<i}, \tilde{\mathbf{a}}_i, \mathbf{s}_i\}$; access to $\tilde{\mathbf{v}}_i$ is controlled by the routed attention introduced in Section III-C.

3) Flow-matching objective: We use flow matching for video and action prediction in both training stages [17]. Within each temporal block, video and actions share a uniformly sampled timestep index τ. For each clean sample x, we sample independent noise $\epsilon \sim \mathcal{N}(0, I)$ and construct

$$x_\tau = (1 - \sigma_\tau)x + \sigma_\tau \epsilon, \qquad u = \epsilon - x, \tag{1}$$

where σ_τ is the noise level and u is the target velocity. We minimize

$$\mathcal{L} = \mathbb{E}[w(\tau)\,\mathrm{MSE_v}(v_\theta^{\mathrm{v}}, u^{\mathrm{v}})] + \mathbb{E}[w(\tau)\,\mathrm{MSE_a}(v_\theta^{\mathrm{a}}, u^{\mathrm{a}})]\,. \tag{2}$$

v_θ denotes the predicted velocity and $w(\tau)$ the timestep weight. Each MSE averages the masked errors for video or actions separately, and the two terms have equal weight.

C. Routed Attention

We introduce routed attention to control whether action queries can read future-video tokens. At inference, *joint* mode denoises future video and actions together, whereas *action-only* mode predicts actions from observed context without generating future video (Fig. 3 (c,d)). Mixed-routing training randomly selects between the two attention routes.

Let p be the probability of using the joint route and draw $z \sim \mathrm{Bernoulli}(p)$. The token groups visible to action queries in block i are

$$\mathcal{K}_i^{\mathrm{a}}(z) = \{\mathbf{c}_{<i}, \tilde{\mathbf{a}}_i, \mathbf{s}_i\} \cup \begin{cases} \{\tilde{\mathbf{v}}_i\}, & z = 1, \\ \varnothing, & z = 0. \end{cases} \tag{3}$$

Only action queries change access to future-video tokens; both routes retain the video-prediction path (Fig. 3 (a,b)).

We use $p = 0.5$ for mixed-routing training and $p = 0$ or $p = 1$ for fixed-route training. The mixed-routing checkpoint supports both joint and action-only inference. Based on the ablations in Section IV-F, we adopt $p = 0.5$ mixed-routing training with action-only inference as our default.

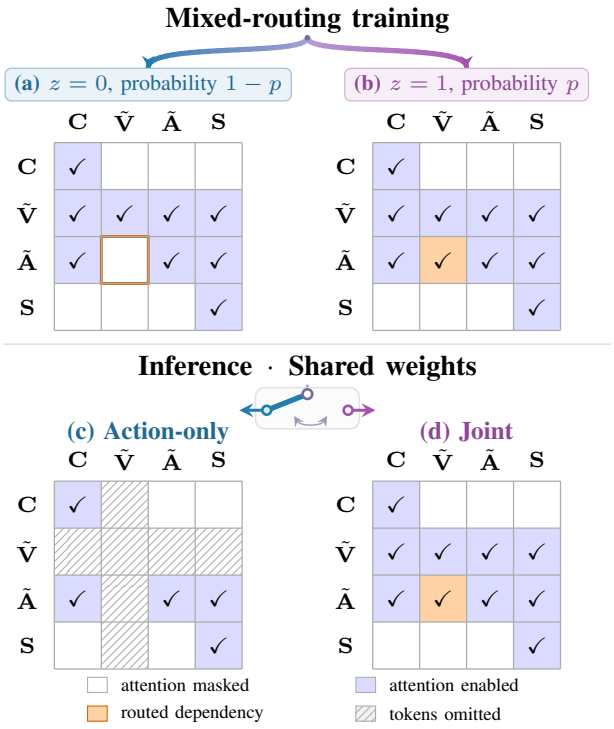


Fig. 3. **Routed attention in training and inference.** (a,b) Mixed-routing training samples the action-only or joint route with probabilities $1 - p$ and p. The same checkpoint supports (c) action-only inference and (d) joint inference. Rows are queries and columns are keys: $\mathbf{C}$ groups preceding clean visual context; $\tilde{\mathbf{V}}$, $\tilde{\mathbf{A}}$, and $\mathbf{S}$ denote noisy future video, noisy actions, and state, respectively. Orange highlights the routed action-to-video dependency (Eq. (3)); hatching marks omitted tokens.

D. Egocentric Action Pretraining

1) Data: We combine approximately 822 hours from EgoDex [3] and 1,166 hours from EgoVerse [18] after preprocessing. EgoDex focuses on dexterous manipulation, while EgoVerse covers everyday activities across varied environments; both provide tracked hand poses. This 2k-hour dataset is used to compare the pretraining targets below.

For larger-scale latent-action pretraining, we augment the 2k-hour dataset with nested 3k- and 8k-hour subsets of EgoWE, yielding 5k- and 10k-hour settings, respectively. EgoWE is our in-house dataset of human egocentric videos covering assembly, crafting, packaging, and other manual activities. Both subsets are sampled to have similar distributions of recording-session durations.

2) Pretraining targets: We compare three objectives on the same 2k-hour pretraining dataset to identify an effective action representation for humanoid transfer.

a) Latent action ($\mathbf{a}_{\text{lat}} \in \mathbb{R}^{32}$): A frozen latent action model (LAM) [9] encodes consecutive frames into $\mathbf{a}_{\text{lat},t} = E_{\text{LAM}}(I_t, I_{t+1})$. The LAM was trained to reconstruct the next frame from the current frame and this code. This visual-transition target captures coarse-grained interaction semantics without prescribing human hand kinematics.

b) Hand pose ($\mathbf{a}_{\text{hand}} \in \mathbb{R}^{48}$): We use tracked hand poses as geometric targets. Each hand contributes a 3D wrist position, a 6D rotation representation [19], and five 3D fingertip positions, expressed in a unified egocentric frame after source-specific coordinate transforms.

c) Action-free video: We replace action targets with zeros and disable the action loss. This adapts the model through video prediction alone while retaining the action-token interface used for robot post-training.

For all three objectives, human-video pretraining starts from Wan2.1 [14] with the same routing probability, followed by the same robot post-training protocol. The *Wan base* reference uses the same initialization but skips human-video pretraining and proceeds directly to robot post-training.

3) Configuration: Human pretraining uses a one-dimensional zero dummy state. We update the full DiT while keeping the VAE, text encoder, image encoder, and LAM frozen. The 10k-hour model is pretrained for 50k steps on 152 H100 GPUs with batch size 4 per device (608 globally), peak learning rate 6×10^{-5}, and a cosine schedule. We use FSDP2 for distributed training.

E. Post-Training on Unitree G1

1) Robot interface: For Unitree G1, $\mathbf{a} \in \mathbb{R}^{80}$ concatenates $[\mathbf{q}_{\text{hand}}, \mathbf{q}_{\text{neck}}, \mathbf{d}_{\text{body}}]$, where $\mathbf{q}_{\text{hand}} \in \mathbb{R}^{14}$ contains joint-position targets for both hands, $\mathbf{q}_{\text{neck}} \in \mathbb{R}^{2}$ contains neck pitch and yaw targets, and $\mathbf{d}_{\text{body}} \in \mathbb{R}^{64}$ is the latent body command for the SONIC controller [20]. The state $\mathbf{s} \in \mathbb{R}^{45}$ concatenates 29 body, 14 hand, and 2 neck joint positions.

We inherit the pretrained weights and reinitialize only the parameters whose shapes change with the robot state and action dimensions. All trainable parameters are then jointly optimized on robot demonstrations.

We jointly post-train a single policy on the nine household tasks in Fig. 4, using mixed-routing training with $p = 0.5$ unless otherwise noted. Post-training runs for 32k steps on eight H100 GPUs with batch size 4 per device (32 globally), peak learning rate 10^{-5} and a cosine schedule.

F. Efficient Inference and Deployment

Continuous whole-body control requires the robot to keep executing actions while the next chunk is predicted. During walking, kneeling, or rising, stale body commands can destabilize motion. Our deployment combines inference acceleration, quantization, and real-time chunking to sustain a continuous 30 Hz action stream (Fig. 2 (d)). SONIC [20] reads the latest body command and robot state in a 50 Hz feedback loop to generate body joint targets $\mathbf{q}_{\text{body}}$.

Acceleration and quantization. Action-only inference removes future-video denoising. At the configured input resolution, each DiT denoising forward pass processes 25 action/state tokens, compared with 465 video/action/state tokens in joint mode, reducing the active sequence length by 94.6%. We cache the keys and values of observed video tokens and reuse them across denoising passes and subsequent chunks. Classifier-free guidance [21] is applied only to video, so the action-only path also skips the unconditional DiT pass. At selected denoising steps, we reuse the latest predicted velocity to further accelerate inference. We use FP8 E4M3 for DiT linear-layer weights and activations, weight-only FP8 for text-encoder linear layers, INT8 for

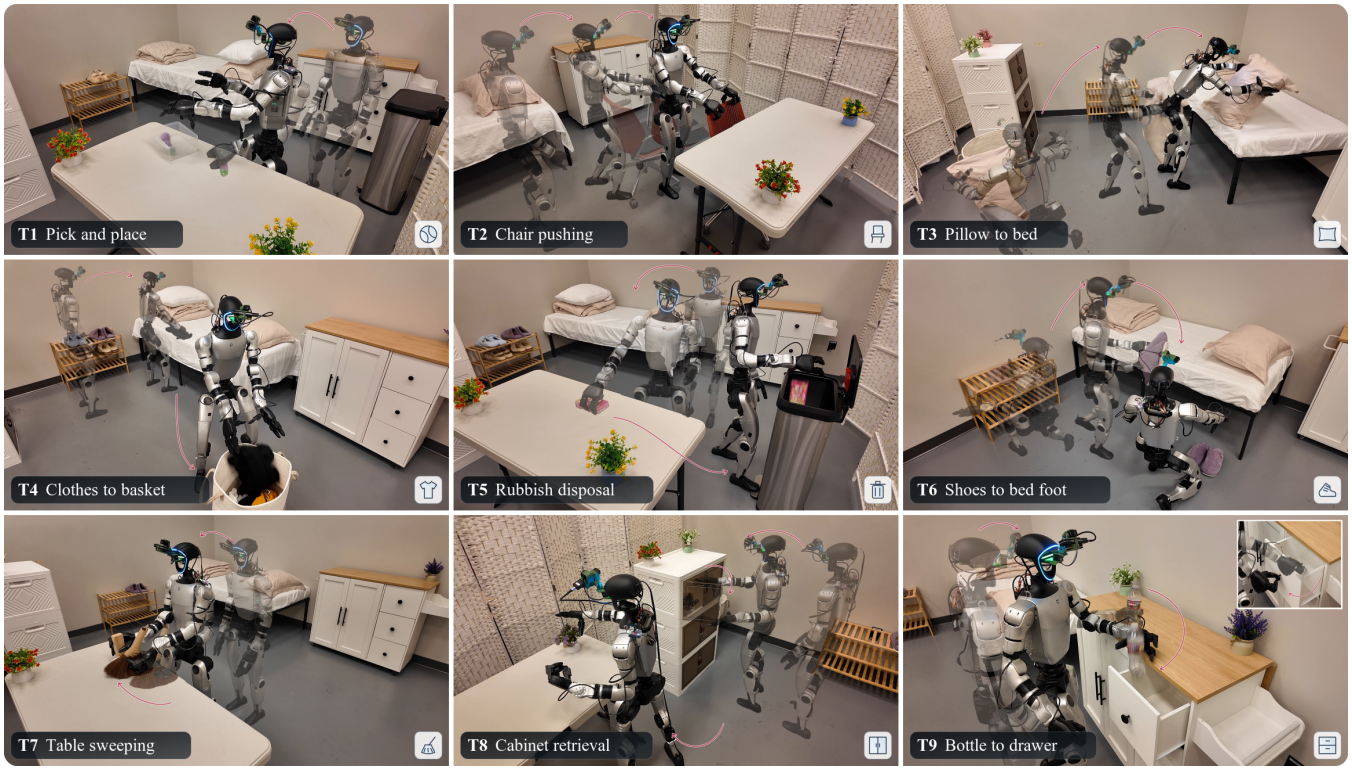


Fig. 4. **Nine in-distribution humanoid tasks.** The task suite covers household manipulation involving whole-body motions such as walking, turning, and kneeling. Overlaid poses and arrows illustrate task progression, with earlier stages shown translucently. Task identifiers T1–T9 are used consistently throughout the in-distribution results. T9 stores bottles in a drawer; its inset shows drawer interaction.

the text-embedding matrix, and FP8 for self-attention KV caches, while attention uses BF16 inputs. Together with blockwise loading, quantization reduces peak allocated GPU memory by approximately 46% relative to the BF16 baseline, supporting deployment on a single RTX 5090.

Real-time chunking. We use real-time chunking (RTC) [13] to overlap prediction and execution, generating $H = 24$ actions and replanning every $S = 12$ actions. Pseudoinverse guidance [22] encourages continuity across chunks using a soft mask that emphasizes the committed prefix and tapers over the remaining overlap. We approximate the denoised-action Jacobian by the identity, avoiding backpropagation through the quantized DiT.

IV. Experiments

We evaluate how egocentric pretraining supports humanoid loco-manipulation, from acquiring demonstrated tasks to handling changed conditions and composing learned skills. We compare pretraining objectives, data scales, and existing policies on in-distribution (ID) tasks, then evaluate seen-task generalization (L1) and compositional generalization (L2) (Section IV-E). A separate ID ablation tests the training and inference choices enabled by routed attention.

A. Experimental Setup

We evaluate on a Unitree G1 equipped with two 7-DOF Dex3-1 dexterous hands, a two-degree-of-freedom TWIST2 neck [27], and a neck-mounted ZED Mini stereo camera. Our robot demonstration dataset contains 4.02 hours covering nine household task families and 53 instruction-defined

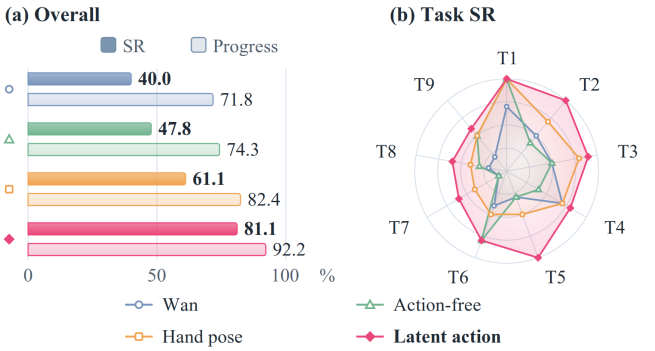


Fig. 5. **Egocentric pretraining objectives.** (a) Overall SR and task progress use darker and lighter bars, respectively; (b) per-task SR uses a 0–100% radial scale and the task identifiers in Fig. 4. The three ego-pretrained models share the same 2k-hour video subset; Wan base receives no additional egocentric pretraining. Each model is evaluated on ten trials per task (N=90), with tasks weighted equally in the overall scores.

subtasks, with 5–10 demonstrations per subtask. These tasks combine manipulation with whole-body motions such as walking, turning, and kneeling (Fig. 4). All compared models are post-trained on this shared dataset and use the same head-camera RGB observations and robot action interface described in Section III-E.

We conduct ten trials per ID task across its subtasks, and ten trials per L1 condition or L2 composition, yielding 90, 60, and 40 trials per model, respectively. Evaluation instructions, objects, and starting configurations are matched across models within each task or condition. L2 uses the ID checkpoints without additional training.

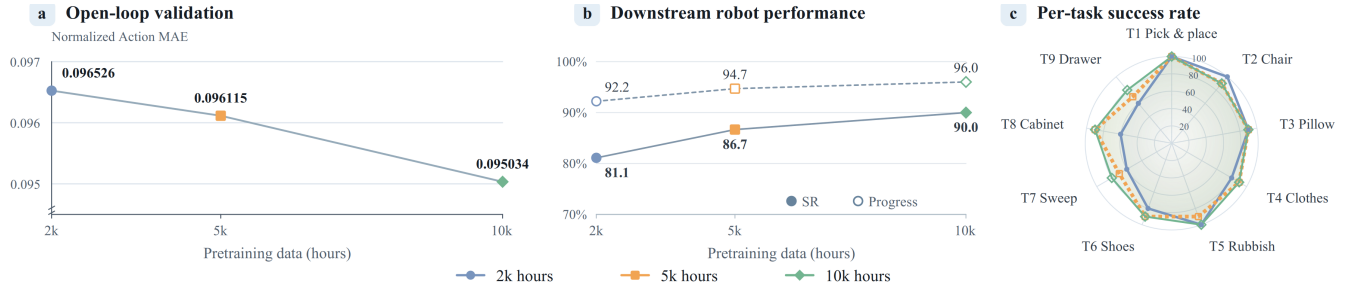


Fig. 6. **Egocentric data scaling.** (a) Normalized action MAE on the robot validation set at the 32k post-training checkpoint. (b) Overall ID SR and progress. (c) Per-task ID SR, with task identifiers from Fig. 4. All three data scales use ten robot trials per task (N=90 per model).

TABLE I: **In-distribution comparison.** Success rate (%) for nine tasks and overall SR/progress. Bold indicates the best score per column.

Category	Method	PnP	Chair	Pillow	Clothes	Rubbish	Shoes	Sweep	Cabinet	Drawer	Avg. SR	Avg. Prog.
WAM	Cosmos3 [23]	80	70	50	60	70	30	30	30	10	47.8	74.3
	DreamZero [1]	70	70	10	70	30	10	0	20	20	33.3	63.5
	Fast-WAM [11]	60	30	0	40	20	20	10	0	0	20.0	56.4
	Dit4dit [24]	60	80	50	30	10	40	0	10	0	31.1	65.1
VLA	Ψ_0 [5]	80	80	70	30	60	30	30	20	10	45.6	70.8
	GR00T N1.7 [25]	60	70	60	60	60	40	20	10	0	42.2	68.6
	$\pi_{0.5}$ [26]	40	20	0	20	10	20	0	10	0	13.3	50.4
Ours	**WE-Dream-10k**	**100**	**90**	**90**	**90**	**100**	**90**	**80**	**90**	**80**	**90.0**	**96.0**

We report success rate (SR), the proportion of trials completing all task stages, and task progress, the fraction of stages completed in each trial. Stage counts are task-specific. For both metrics, we average over trials within each task or condition then average equally across tasks or conditions.

B. Latent-Action Pretraining Improves Humanoid Transfer

Which supervision target best transfers egocentric video knowledge to humanoid loco-manipulation? We compare action-free, hand-pose, and latent-action pretraining on the same 2k-hour dataset with the same initialization. The three pretraining objectives use the same update counts and effective batch sizes. Wan base provides a reference without egocentric pretraining. All four models share the robot post-training and action-only inference protocol.

Latent-action pretraining achieves 81.1% overall SR, exceeding hand-pose supervision by 20.0 percentage points and outperforming both action-free pretraining and Wan base (Fig. 5). Its advantage over hand-pose supervision extends to eight of the nine tasks.

The differences are particularly evident in tasks requiring coordinated whole-body motion. In our observations, the hand-pose-pretrained model failed to lift its foot sufficiently to operate the rubbish-bin pedal and lost balance while kneeling to retrieve shoes. These failures involve lower-body motion and postural stability, beyond hand manipulation alone. Latent actions derived from visual transitions may provide richer supervision for such coordinated motion than hand-centric pose targets, offering a possible explanation for the observed transfer advantage.

C. Scaling Pretraining Improves Downstream Performance

We examine how scaling latent-action pretraining from 2k to 10k hours affects downstream humanoid performance. We evaluate models pretrained on 2k, 5k, and 10k hours of human video using the same robot post-training protocol and demonstration dataset. For each data scale, robot post-training starts from the final 50k-step pretraining checkpoint.

In open-loop evaluation on the robot validation set, normalized action MAE decreases across the three scales, with a 1.55% relative reduction from 2k to 10k hours (Fig. 6 (a)). The benefit also appears in closed-loop execution: overall ID success rises from 81.1% to 90.0%, an improvement of 8.9 percentage points, while task progress increases from 92.2% to 96.0% (Fig. 6 (b)).

Task success improves notably in cabinet retrieval, tabletop sweeping, and drawer storage (Fig. 6 (c)). In observed rollouts, the 10k model opens cabinet doors and drawers more accurately and performs smoother, more continuous sweeping motions than the 2k model.

D. In-Distribution Comparison with Baselines

How does WE-Dream compare with existing WAMs and VLAs on humanoid task execution? We evaluate four WAMs, Cosmos3 [23], DreamZero [1], Fast-WAM [11], and Dit4dit [24], and three VLAs, Ψ_0 [5], GR00T N1.7 [25], and $\pi_{0.5}$ [26], under the shared protocol in Section IV-A. RTC follows each method's open-source implementation.

WE-Dream-10k achieves the highest SR on all nine tasks (Table I). Its overall SR of 90.0% exceeds the strongest WAM baseline, Cosmos3, by 42.2 percentage points and the strongest VLA baseline, Ψ_0, by 44.4 points. Task progress

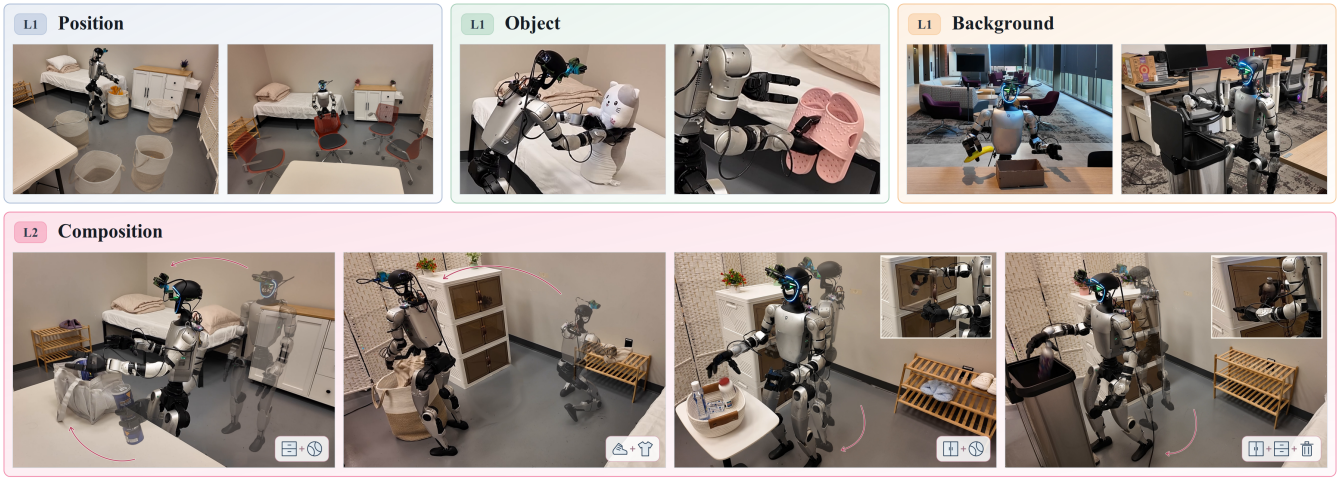


Fig. 7. **Generalization task illustrations.** Top row (L1), left to right: laundry-basket and chair position changes, pillow and shoe replacements, and pick-and-place and rubbish disposal in new backgrounds. Bottom row (L2), left to right: pack chips, collect clothes, organize medicine, and discard a bottle. Position panels overlay different object placements; object and background panels use single frames. L2 temporal overlays show earlier poses translucently and the final pose opaquely; the insets show cabinet interaction. Corner icons identify the contributing ID tasks.

(a) L1: Seen-task generalization

	Position		Object		Background		Overall	
	Clothes	Chair	Pillow	Shoes	PnP	Rubbish	SR	Progress
Cosmos3	50	70	30	20	10	30	35.0	64.6
DreamZero	40	50	10	10	10	20	23.3	54.9
Ψ_0	40	40	10	50	10	20	28.3	62.3
GR00T N1.7	40	30	30	20	10	20	25.0	63.2
Wan base	30	50	20	20	10	0	21.7	61.4
Action-free-2k	30	40	20	40	30	0	26.7	55.1
Hand pose-2k	50	60	20	20	40	30	36.7	66.3
Latent-action-2k	90	**80**	60	60	90	40	70.0	85.5
WE-Dream-5k	80	**80**	50	80	**100**	50	73.3	89.3
WE-Dream-10k	**100**	**80**	**80**	**90**	**100**	**70**	**86.7**	**93.9**

(b) L2: Compositional generalization 0 100%

	Composition				Overall	
	Pack chips	Clothes	Medicine	Bottle	SR	Progress
Cosmos3	0	0	10	0	2.5	30.4
DreamZero	20	0	10	0	7.5	34.6
Ψ_0	0	0	0	0	0.0	23.8
GR00T N1.7	0	0	0	0	0.0	22.5
Wan base	0	0	10	0	2.5	28.8
Action-free-2k	10	0	10	0	5.0	35.7
Hand pose-2k	50	20	20	0	22.5	55.8
Latent-action-2k	80	50	60	**60**	62.5	78.7
WE-Dream-5k	80	**60**	70	50	65.0	82.9
WE-Dream-10k	**90**	**60**	**80**	**60**	**72.5**	**84.6**

Fig. 8. **Generalization results.** L1 changes position, object, or background within learned tasks; L2 composes skills across tasks without additional training. Task cells report SR (%) from ten trials per condition or task; darker shading indicates higher SR on a shared color scale. Overall SR and progress weight conditions or tasks equally (N=60 for L1 and N=40 for L2). Bold values indicate the best score in each column, including ties.

also leads at 96.0%. The largest margins over the best baseline for each task occur in cabinet retrieval and storing bottles in drawers, both at 60 percentage points.

Across multiple baselines, we repeatedly observed imprecise manipulation, unstable motion, and stalls. In contrast, WE-Dream completes multi-stage tasks more reliably.

E. Generalization to New Conditions and Skill Compositions

Our generalization benchmark covers changed task conditions (L1) and unseen skill compositions (L2), illustrated in Fig. 7.

1) L1: seen-task generalization: Can learned tasks transfer to new positions, objects, and backgrounds? L1 evaluates six conditions: changing laundry-basket and chair positions; replacing pillows with a green pillow and a long body pillow, and shoes with pink and white pairs; and moving pick-and-place to a lounge and rubbish disposal to an office outside the training room.

WE-Dream-10k achieves the highest overall L1 SR and task progress among the evaluated models, at 86.7% and 93.9%, respectively (Fig. 8 (a)). Its SR exceeds the strongest external baseline, Cosmos3, by 51.7 percentage points.

Unlike baselines that often follow ID-like trajectories, WE-Dream adapts across all three L1 categories. Under position changes in clothes collection, it walks toward a laundry basket positioned at a previously unseen angle. Under object changes in pillow arrangement, it brings both hands farther inward to securely grasp a narrower pillow. Under background changes in pick-and-place and rubbish disposal, it successfully executes tasks in unfamiliar environments.

2) L2: compositional generalization: Can separately learned skills be composed into unseen task sequences? L2 evaluates four compositions of skills from two or three robot post-training tasks, using manipulated objects unseen in the robot demonstrations:

Pack chips: combine horizontal grasping from drawer storage with container placement from pick and place.

Collect clothes: combine kneeling from shoe retrieval with cloth handling and basket placement from clothes collection.

Organize medicine: combine cabinet opening and retrieval with container placement from pick and place.

Discard a bottle: combine cabinet opening and retrieval, lateral bottle grasping from drawer storage, and pedal-operated

TABLE II: **Routed-attention ablation.** Overall ID SR and progress (%). p is the training joint-route probability; highlighted rows share a checkpoint.

Train p	Inference	SR (%)	Progress (%)
0	*action-only*	25.6	60.8
0.5	*action-only*	**40.0**	**71.8**
0.5	*joint*	38.9	68.9
1	*joint*	31.1	65.1

disposal from rubbish disposal.

WE-Dream-10k achieves the highest L2 SR and task progress, at 72.5% and 84.6%, respectively (Fig. 8 (b)), compared with 7.5% and 34.6% for the strongest external baseline, DreamZero.

Baselines tend to follow familiar task trajectories rather than recombine skills for the new instruction and visual context. In chip packing, they repeat the vertical pick-and-place grasp instead of grasping the cylindrical container horizontally. They also revert to shoe-hooking after kneeling to collect clothes, or continue toward the table instead of stepping on the trash-bin pedal when discarding a bottle. In contrast, WE-Dream more reliably combines learned skills into the requested task sequences.

Across L1 and L2, latent-action pretraining outperforms action-free and hand-pose in both SR and task progress at the same 2k-hour scale. Both metrics improve further as latent-action pretraining scales to 5k and 10k hours.

F. Routed-Attention Ablation

Does mixed-routing training improve task performance while enabling faster *action-only* inference? To avoid bias from the $p = 0.5$ routing used during egocentric pretraining, we initialize all ablations directly from Wan2.1 and post-train on the same nine-task robot dataset for the same number of updates, using $p \in \{0, 0.5, 1\}$. RTC is enabled for *action-only* inference and disabled for *joint* inference.

Mixed-routing training improves both inference modes over their fixed-route counterparts, increasing SR by 14.4 percentage points for *action-only* and 7.8 points for *joint* inference (Table II).

With the same mixed-routing checkpoint, *action-only* achieves similar SR to *joint* inference (40.0% versus 38.9%), with no observed decrease in aggregate task progress. On an RTX 5090, it reduces mean server-side latency from 751 ms to 260 ms, a 2.9× speedup. We therefore adopt mixed-routing training with *action-only* inference as our default.

V. Conclusion

We presented WE-Dream, a humanoid world-action foundation model that combines latent-action pretraining with routed attention. Evaluations on our benchmark show stronger transfer from latent-action pretraining across demonstrated tasks, changed conditions, and unseen skill compositions, with further gains from more human video. Our ID ablation supports mixed-routing training with action-only deployment, combining higher success than fixed-route training with lower inference cost than joint inference. With FP8 quantization and real-time chunking, the 16B-parameter policy runs on a single RTX 5090 with ∼260 ms latency and continuous 30 Hz whole-body action execution.

Our findings are limited to a humanoid with Dex3-1 hands and require validation across other embodiments and hand morphologies. Resource constraints also limit the scale of our real-robot data and evaluation.

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[TODO: Check the final ICRA 2027 page and supplementary-material rules before submission, then decide which implementation details remain in the main PDF.]

APPENDIX I
IMPLEMENTATION DETAILS

A. Pretraining Datasets

1) EgoVerse: EgoVerse [18] aggregates four capture sources, summarized in Table III.

TABLE III: EgoVerse dataset breakdown by data source.

Source	Episodes	Frames	Hours
MECKA (bimanual)	41,612	89,233,982	826.2
Aria (bimanual + left_arm + right_arm)	2,420	11,240,739	104.1
Scale	17,090	23,389,757	216.6
Eva (bimanual + right_arm)	3,031	3,342,095	30.9
Total	**64,153**	**127,206,573**	**1177.8**

a) Low-quality episode filtering: (1) **Eva source**: all Eva episodes are excluded (robot joint data without hand tracking). (2) **Debug/test recordings**: MECKA episodes with `task_text="debug"`, Aria episodes with `task_text` set to "N/A" or "test". (3) **Short episodes**: anything below 50 frames (∼1.7 s at 30 Hz). Table IV reports what each filter removes, and Table V the resulting dataset. Latent-action subsets drop each episode's final frame (a latent code needs an adjacent pair) and therefore contain marginally fewer frames.

TABLE IV: EgoVerse episodes removed during preprocessing

Reason	Episodes	Frames	Hours
Eva source (*no hand tracking*)	3,031	3,342,095	30.9
MECKA debug recordings	7,014	1,220,439	11.3
Aria N/A annotation	2	6,539	0.1
Aria test recording	1	206	<0.1
Eva test recording	1	405	<0.1
MECKA short clips (also debug)	2	30	<0.1
Eva short clips	1	45	<0.1

TABLE V: EgoVerse dataset after preprocessing, grouped by resolution subset and action representation.

	Source	Action	Episodes	Frames	Hours
360p	MECKA	Hand pose	34,598	88,013,543	814.9
480p	Aria + Scale	Hand pose	19,507	34,623,751	320.6
Hand pose total			**54,105**	**122,637,294**	**1135.5**
360p	MECKA	Latent	34,598	87,978,945	814.6
480p	Aria + Scale + Eva	Latent	22,536	37,942,860	351.3
Latent total			**57,134**	**125,921,805**	**1166.0**

b) Instruction enrichment: The Scale source stores terse task descriptions (e.g. "Sort utensils") but also carries step-level narration. We concatenate the two, producing fuller instructions.

c) Hand-pose target processing: The three datasets do not share a reference frame. EgoDex's ARKit-origin poses are transformed into the camera frame; EgoVerse gives SLAM world coordinates that we transform into the head/device frame; the in-house dataset gives a pelvis-referenced skeleton that we transform into the camera frame via per-frame extrinsics. Additional steps:

1) *Missing-hand zero padding.* When a hand's source keypoints are identically zero, we skip the transform and write zeros directly in the camera frame to avoid spurious offsets.
2) *Position outlier removal.* If $\max(|\mathbf{p}|) > 2.0\,\mathrm{m}$ for any hand in a frame, that hand's 24-D vector is zeroed.
3) *Wrist convention alignment.* A per-source rotation aligns all wrist `rot6d` values to the EgoDex (ARKit) convention.

2) EgoDex: EgoDex [3] contributes 338,234 ARKit-captured 30 Hz episodes. Only the short-episode filter applies, removing 12,456 episodes (3.7%).

TABLE VI: EgoDex dataset statistics before and after preprocessing.

	Action	Episodes	Frames	Hours
Raw	–	338,234	89,580,126	829.4
Processed	Hand pose	325,778	89,062,283	824.7
Processed	Latent	325,778	88,736,505	821.6

3) In-House Egocentric Dataset: **[TODO: Write the full description of the in-house dataset: capture hardware, segment-level statistics, anonymization, and any details moved from the method section.]**

4) The scaling ladder: Table VII summarizes the three pretraining data scales.

TABLE VII: **The egocentric pretraining ladder.** The EgoWE subsets are nested with matched stratum proportions. The EgoDex + EgoVerse base is identical in all three, so the public/in-house source proportions change with scale.

Rung	Base frames	EgoWE frames	Total frames	Hours
2k	214,658,310	–	214,658,310	2,000
5k	214,658,310	329,347,920	544,006,230	5,037
10k	214,658,310	869,375,293	1,084,033,603	10,037

B. Latent Action Model

We use DreamDojo's [9] released `LAM_400k.ckpt` checkpoint, keeping the LAM frozen as a feature extractor for latent-action targets.

a) Architecture: A 700M-parameter VAE built on the spatiotemporal transformer of Genie [8], with 24-block encoder and decoder:

$$\mathbf{a}_{\text{lat}} = E_{\text{LAM}}(I_t, I_{t+1}), \qquad \hat{I}_{t+1} = D_{\text{LAM}}(\mathbf{a}_{\text{lat}}, I_t), \quad (4)$$

where E_{LAM} maps a consecutive frame pair to $\mathbf{a}_{\text{lat}} \in \mathbb{R}^{32}$ and D_{LAM} reconstructs I_{t+1} from $\mathbf{a}_{\text{lat}}$ and I_t. Training uses the standard VAE objective with $\beta = 10^{-6}$.

b) Extraction: Frames are resized to 320×240; extraction uses stride 1 (adjacent frames at 30 Hz). Episodes shorter than 50 frames are discarded and each episode's tail is dropped where a pair cannot be formed (∼2% of rows). Normalization statistics are recomputed per dataset.

C. Model and Training

1) Architecture: The backbone is Wan2.1-I2V-14B-480P [14], with 40 DiT blocks, model width 5120, and 40 attention heads. Trainable action and state encoders map their inputs into the model dimension, with per-stream timestep modulation for noise conditioning. A video head and an action decoder map the shared features to flow velocities;

state tokens provide conditioning without a prediction head. Video is processed at 320×176. For a 33-frame training clip, the VAE compresses 8× spatially and 4× temporally into nine latent frames; a 1×2×2 patchifier then produces 220 tokens per latent frame. The latent sequence contains one reference frame followed by four two-frame blocks, each aligned with one state token and 24 action tokens. Shorter samples use fewer valid blocks; padding is excluded from the loss. Rotary embeddings use separate frequency banks for action and state groups so that a chunk's phase tracks the latent frame it belongs to, allowing the KV cache to be reused across autoregressive steps.

a) Token layout and I2V conditioning: Teacher-forced sequences are stored as [clean video]‖[noisy video]‖[action]‖[state]; logical temporal-block alignment is imposed by the attention mask rather than by interleaving modalities. The Wan I2V condition concatenates a visibility mask and a VAE encoding of the first frame with zero-padded future slots to both clean and noisy video inputs. The CLIP image embedding is prepended to the text context used by cross-attention.

2) Training configuration: Table VIII lists the hyperparameters for 10k-hour pretraining and robot post-training.

TABLE VIII: **Training hyperparameters.** Both stages train every backbone parameter; the VAE, text encoder, CLIP encoder and LAM stay frozen. The pretraining column reports the 10k-hour run.

	Pretraining	**Post-training**
Initialization	Wan2.1-I2V-14B	pretrained
GPUs (H100 80GB)	152	8
Per-device batch	4	4
Global batch	608	32
Steps	50,000	32,000
Peak LR	6×10^{-5}	1×10^{-5}
Schedule	cosine	cosine
Warmup ratio	0.03	0.05
Weight decay	0.01	1×10^{-5}
Precision	bf16 (tf32 matmul)	bf16 (tf32 matmul)
Sharding	FSDP2	FSDP2
Resolution	320 × 176	320 × 176
Action horizon H	24	24
Chunks per clip	4	4
$\dim(\mathbf{a})$	32 (latent) / 48 (hand)	80
$\dim(\mathbf{s})$	1 (placeholder)	45
Routing p	0.5	0.5

a) Noise coupling: For each future temporal block, the default configuration samples one uniform timestep index shared by its two video latents; all action tokens in the corresponding chunk inherit that index. The reference latent uses an independently sampled index. Both losses use the scheduler's non-uniform timestep weighting.

b) Distributed training: Each data-parallel rank independently samples one attention route per microbatch. Using the same route on all GPUs in each training step led to unstable training and substantial performance variation across checkpoints. Training uses PyTorch FSDP2 for parameter and optimizer-state sharding. Two accommodations are required at this model and sequence scale. First, the dataloader pads every sequence in a batch to a uniform number of chunks, and a per-sample validity mask removes the padded latent frames and action chunks from both loss terms. Second, the routed mask interacts with compiled attention: the `action_attend_to_noisy_video` flag flips between steps, causing the compiler to specialize on it and record a second CUDA graph whose private memory pool is never released. We therefore leave attention compilation off during training (at ∼4% step-time cost) and enable it only at inference where the flag is fixed.

3) Inference:

a) Autoregressive rollout: The first call sends a single frame; the model encodes it with CLIP as the global condition and with the VAE into one latent, seeds the KV cache, and denoises the first chunk. Subsequent calls send 4 frames sampled at fixed offsets from the 24-frame execution buffer; these are re-encoded into latents and appended to the cache, and the next chunk is denoised conditioned on the cache and the current state $\mathbf{s}_t$. When the cache reaches its trained capacity of $L_{\max}$=9 latent frames, it is reset from the most recent frame, keeping the rotary phase inside the trained range. The UniPC scheduler performs 16 numerical updates per action chunk. With `NUM_DIT_STEPS=4`, only updates 0, 1, 7, and 12 run a DiT forward pass; the other updates reuse the latest predicted velocity. Thus, this setting uses four network evaluations within a 16-update denoising schedule. **[TODO: Confirm the base schedule and network-evaluation counts used for ablations and final deployment.]**

b) Denoising token count: At 320×176 resolution, spatial VAE compression by 8 and a 2×2 spatial patch produce $(320/16)(176/16) = 220$ tokens per video latent. A joint denoising pass therefore processes $2(220) + 24 + 1 = 465$ tokens; an action-only pass processes $24{+}1 = 25$. The 94.6% reduction concerns tokens updated through the DiT per pass. Cached observation keys and values remain available to attention, and observation encoding is performed separately. **[TODO: Remember to check the numbers here]**

c) Compilation and warmup: Five modules are compiled under `reduce-overhead` mode: the joint and action-only DiT paths, the text encoder, the CLIP encoder and the VAE encoder. Since the KV cache grows per step and compiled execution specializes on its shape, a one-time warmup walks the full autoregressive cycle for the selected inference mode(s). Under real-time chunking, every generation phase must also be pre-recorded. Without this warmup, a previously unseen shape introduces a long compilation pause into the command stream. **[TODO: Confirm the cold-start delay, warmup call count, and wall-clock warmup time on the final deployment GPU.]**

d) CFG-free action-only inference: Joint denoising uses classifier-free guidance [21] to sharpen the generated video. Guidance is applied to the video velocity only; the action velocity always uses the conditional prediction alone. In *action-only* mode no video is integrated, so the unconditional branch has no consumer and is dropped together with its text encoding, KV cache and DiT forward pass. **[TODO: Confirm the classifier-free guidance scale used in all reported runs.]**

e) Quantization: The DiT is quantized layer-wise during loading: linear layers inside the transformer blocks use FP8 E4M3 with dynamic per-tensor activation scaling. Patch embeddings, output heads, normalization, and attention kernels retain their original precision. The text encoder's transformer linear weights use FP8 weight-only quantization; its token embedding uses symmetric INT8 storage with a scale per vocabulary entry. Self-attention keys and values are stored in FP8 E4M3 with per-head scales and dequantized to bf16 for attention; cross-attention caches remain in bf16. Quantization runs block-by-block during loading to limit the transient GPU-memory peak. Together with blockwise loading, quantization reduces peak allocated GPU memory by approximately 46% relative to the BF16 baseline. **[TODO: Report the BF16 and quantized peak allocated GPU memory in GiB and specify whether the peak occurs during loading or inference. Record the measurement configuration and break down memory use by model weights, activations, and KV caches; distinguish allocated from reserved memory.]**

f) Inference latency: On an RTX 5090, mean server-side latency is 260 ms for action-only inference and 751 ms for joint inference, a 2.9× speedup (Section IV-F). **[TODO: Document the latency measurement: checkpoint, input resolution, context length, action horizon, denoising schedule and DiT evaluations, precision, caching, compilation, and RTC settings for each mode. Specify whether timing includes observation encoding, cache resets, and RTC guidance; report warmup exclusions and the number of timed calls.]**

g) RTC guidance: Let A^{prev} be the previous normalized chunk, $H = 24$ the prediction horizon, and $S = 12$ the execution stride. The overlap target is $Y_j = A^{\text{prev}}_{j+S}$ for $0 \le j < H - S$; the remaining entries have zero guidance weight. At noise level σ, estimate the clean action as $\hat{A} = X_\sigma - \sigma v_\theta$ and modify the velocity by

$$v_{\text{RTC}} = v_\theta - g(\sigma)\, W \odot (Y - \hat{A}), \tag{5}$$

where, for $0 < \sigma < 1$,

$$g(\sigma) = \min\left(g_{\max}, \frac{\sigma^2 + (1-\sigma)^2}{\sigma(1-\sigma)}\right). \tag{6}$$

The implementation explicitly handles the singular coefficient at $\sigma = 1$ using the guidance cap. For a predicted inference delay of d executed ticks, $W_j = 1$ for $j < d$, $W_j = 0$ for $j \ge H - S$, and $W_j = c_j(e^{c_j} - 1)/(e - 1)$ otherwise, where $c_j = (H-S-j)/(H-S-d+1)$. This is an identity-Jacobian approximation to pseudo-inverse guidance, not hard clamping of the predicted prefix.

h) RTC observation updates and delay estimates: Training groups visual latents as one reference followed by two-latent blocks $(1 + 2 + 2 + \cdots)$. In the stride-one RTC variant, each 12-action execution window adds one observed latent to the cache $(1 + 1 + 1 + \cdots)$, while the policy still predicts 24 actions per replan. For new frames I_1, I_2, the VAE receives $[I_1, I_1, I_1, I_2, I_2]$: two copies of each frame plus a prepended boundary frame. It returns two latents, of which only the final content latent is cached. Compared with training, one-at-a-time cache writes remove within-block access to the second latent. Replanning also visits even latent start positions, whereas training aligns action chunks with odd starts; this shifts the relative visual/action rotary alignment by one latent. These changes concern context encoding and temporal alignment, not the 24-action prediction horizon. The client estimates inference delay in executed action ticks to set the guided prefix length. A separate delay history for cache-reset steps lets the client anticipate their extra encoding cost without increasing the guided prefix on every ordinary step. Fixed-instruction text embeddings are cached and refreshed when the instruction changes or the policy resets. **[TODO: Confirm stride-one use, guidance cap, and delay-estimation settings for the reported real-robot runs; validate the temporal-alignment approximation.]**

i) Blackwell toolchain: Deploying on a consumer Blackwell card required three unrelated fixes: the compiler emitted instructions unsupported by the consumer part, the inductor's symbolic simplification overflowed the default C stack, and a debug watchdog faulted during long compilations. We record this because it is the kind of failure that makes a result irreproducible for reasons unrelated to the method.

j) Inter-chunk holding: When real-time chunking is disabled, the client holds hand and neck targets at their last predicted values during the inference gap. The body token requires special handling: holding a stale $\mathbf{d}_{\text{body}}$ can command a pose the controller cannot support. We select a *stable token* — the codebook entry closest to the last predicted body token that decodes to the smallest joint-velocity magnitude — so the robot holds a near-static pose. **[TODO: Complete and verify the stable-token fallback.]**

D. Benchmark Specification

a) Room and objects: **[TODO: Complete the room and object inventory, distinguishing training objects from the L1 replacements in Section IV-E. Task photographs are provided in Fig. 4.]**

b) Primitive vocabulary: Instructions use a small set of verbs, each denoting a distinct grasp or contact strategy: *pick up* (top-down grasp on rigid object), *grasp* (lateral grasp on cylinder), *gather up* (compliant two-handed grasp on cloth), *scoop up* (two-armed lift of pillow), and *hook* (finger-hook extraction of shoe). Keeping these distinct enables L2 to cross a grasp primitive with a transport primitive that were never seen together.

c) Evaluation protocol: Each task or condition has ten planned trials, with SR and progress computed as specified in Section IV-A. ID trials are distributed across subtasks, with evaluation instructions, objects, and starting configurations matched across models. L1 position conditions use ten laundry-basket locations for clothes collection and five chair locations, each tested from near and far robot starting positions. Each object-replacement condition uses five trials per replacement: a green pillow and a long body pillow for the pillow task, and pink and white shoe pairs for the

TABLE IX: **Robot post-training task suite.** Subtask counts and durations describe the robot dataset shown in Fig. 1. Subtasks are defined by distinct instructions, not individual demonstrations. The right column lists constituent actions; hours are rounded independently.

	Task	Subtasks	Hours	Primitives
T1	Pick and place	4	0.20	pick up, place
T2	Push chair	3	0.19	grasp, push
T3	Pillow to bed head	8	0.54	kneel, scoop up, place
T4	Clothes to basket	8	0.62	gather up, drop into basket
T5	Throw rubbish	8	0.28	pick up, turn, pedal-open, throw
T6	Shoes to bed foot	4	0.41	kneel, hook, place
T7	Sweep table	6	0.71	bimanual sweep
T8	Cabinet retrieval	8	0.50	open, grasp, turn, place
T9	Bottle to drawer	4	0.57	grasp, open, place, close
	Total	53	4.02	

shoe task. Recovery attempts are autonomous. A trial is terminated when the on-site evaluator judges that the robot cannot continue; subsequent uncompleted stages receive zero credit. **[TODO: Document the actual reset procedure, lighting controls, trial ordering, scorer identity, and invalid-trial or retry rules. Confirm whether ordering was randomized or scoring was blind; do not assume either from the score records.]**

d) Compositional design: Each L2 task combines skills from two or three robot post-training tasks, including grasping, posture changes, furniture access, and placement. The manipulated objects are not present in the robot post-training demonstrations. Bottle disposal combines cabinet access and retrieval, lateral bottle grasping from drawer storage, and foot-actuated bin opening and release from rubbish disposal. Section IV-E describes the intended new transitions for each case. Object replacement alone is not the criterion for compositional novelty. **[TODO: Add a per-case table listing the source tasks, object, target, and the origin of each recombined primitive. Verify the medicine case against the training trajectories to distinguish a new transition from object replacement.]**

E. Robot Post-Training Data

a) Design: Tasks were selected for coverage of the action space rather than repeated demonstrations of a single behaviour: the suite spans standing, turning, kneeling and bimanual operation. Table IX reports each task family's subtask count and demonstration duration.

b) Collection: **[TODO: Describe the teleoperation setup and retargeting, the number of operators, session structure, how failed demonstrations were filtered, and the dataset version used for reported runs.]**

Appendix II
Additional Results

A. Full Tables

a) In-distribution task progress: Table X reports the per-task progress values for the seven external baselines, four models from the pretraining comparison, routing comparisons, and the 5k and 10k models. The Wan base row is shared by the objective and routing comparisons; it is not an additional evaluation. Task SRs appear in the main-text figures and baseline table.

b) Generalization task progress: Table XI gives per-condition L1 and per-task L2 progress, complementing the success rates in Fig. 8. All reported condition scores use ten trials.

B. Per-Task Analysis

[TODO: For each generalization level, describe which tasks each method succeeds and fails on, and any pattern (fine-grained dexterity versus coarse whole-body motion, short versus long horizon).]

C. Qualitative Rollouts

[TODO: Add representative frame strips from one success and one failure per level.]

TABLE X: **In-distribution task progress (%).** Each task averages ten trials; the overall column weights all nine tasks equally (N=90 per row). Progress is the fraction of completed stages, not a success rate. Wan base is the checkpoint trained with mixed routing and evaluated with action-only inference. Bold values indicate the best progress in each column, including ties.

Method	**PnP**	**Chair**	**Pillow**	**Clothes**	**Rubbish**	**Shoes**	**Sweep**	**Cabinet**	**Drawer**	**Avg.**
Cosmos3	86.7	76.7	84.0	80.0	87.5	66.7	70.0	65.0	52.0	74.3
DreamZero	83.3	83.3	60.0	92.5	60.0	46.7	35.0	58.3	52.0	63.5
Fast-WAM	76.7	56.7	60.0	65.0	60.0	53.3	47.5	46.7	42.0	56.4
Dit4dit	76.7	83.3	76.0	70.0	52.5	70.0	40.0	71.7	46.0	65.1
Ψ_0	86.7	90.0	84.0	75.0	80.0	60.0	62.5	55.0	44.0	70.8
GR00T N1.7	76.7	76.7	84.0	90.0	77.5	70.0	50.0	56.7	36.0	68.6
$\pi_{0.5}$	70.0	46.7	56.0	65.0	47.5	60.0	30.0	38.3	40.0	50.4
Wan base	86.7	73.3	72.0	90.0	60.0	76.7	67.5	61.7	58.0	71.8
Action-free-2k	**100.0**	60.0	76.0	77.5	60.0	91.7	67.5	60.0	76.0	74.3
Hand pose-2k	**100.0**	80.0	92.0	82.5	75.0	80.0	75.0	83.3	74.0	82.4
Latent-action-2k	**100.0**	**100.0**	96.0	95.0	**100.0**	91.7	85.0	88.3	74.0	92.2
Wan p=0, action-only	90.0	60.0	60.0	77.5	57.5	55.0	57.5	41.7	48.0	60.8
Wan p=0.5, joint	**100.0**	63.3	86.0	62.5	60.0	65.0	50.0	63.3	70.0	68.9
Wan p=1, joint	90.0	60.0	54.0	80.0	62.5	68.3	55.0	60.0	56.0	65.1
WE-Dream-5k	**100.0**	93.3	**98.0**	**97.5**	95.0	**96.7**	87.5	**98.3**	86.0	94.7
WE-Dream-10k	**100.0**	93.3	96.0	**97.5**	**100.0**	93.3	**95.0**	95.0	**94.0**	**96.0**

TABLE XI: **Generalization task progress (%).** Each reported condition or task averages ten trials. L1 columns follow position (clothes, chair), object (pillow, shoes), and background (PnP, rubbish) shifts. Overall progress weights conditions or tasks equally within each level (N=60 for L1 and N=40 for L2). Bold values indicate the best progress in each column, including ties.

	L1							**L2**				
Method	**Clothes**	**Chair**	**Pillow**	**Shoes**	**PnP**	**Rubbish**	**Avg.**	**Chips**	**Clothes**	**Medicine**	**Bottle**	**Avg.**
Cosmos3	77.5	80.0	66.0	48.3	53.3	62.5	64.6	36.7	40.0	28.3	16.7	30.4
DreamZero	70.0	70.0	58.0	55.0	26.7	50.0	54.9	46.7	20.0	31.7	40.0	34.6
Ψ_0	75.0	63.3	54.0	75.0	46.7	60.0	62.3	36.7	20.0	20.0	18.3	23.8
GR00T N1.7	72.5	66.7	70.0	63.3	46.7	60.0	63.2	33.3	20.0	20.0	16.7	22.5
Wan base	72.5	70.0	56.0	60.0	60.0	50.0	61.4	36.7	20.0	41.7	16.7	28.8
Action-free-2k	72.5	56.7	48.0	70.0	43.3	40.0	55.1	43.3	36.0	36.7	26.7	35.7
Hand pose-2k	85.0	76.7	46.0	60.0	70.0	60.0	66.3	70.0	48.0	55.0	50.0	55.8
Latent-action-2k	95.0	86.7	88.0	80.0	93.3	70.0	85.5	90.0	68.0	80.0	76.7	78.7
WE-Dream-5k	95.0	**90.0**	86.0	90.0	**100.0**	75.0	89.3	93.3	**70.0**	85.0	**83.3**	82.9
WE-Dream-10k	**100.0**	**90.0**	**92.0**	**96.7**	**100.0**	**85.0**	**93.9**	**96.7**	**70.0**	**90.0**	81.7	**84.6**